

7.24

```
MariaDB [(none)]> create database EmpDepartDB;  
Query OK, 1 row affected (0.009 sec)
```

```
MariaDB [(none)]> use EmpDepartDB;  
Database changed
```

```
MariaDB [EmpDepartDB]> create table Employee(empNo varchar(10) primary key,  
fName varchar(50) not null, lName varchar(50) not null, address varchar(120),  
DOB date not null, sex char(1) check(sex in ('M', 'F')), emPposition varchar(20)  
check(emPposition in ('Manager', 'Team Leader', 'Analysts', 'Software  
Developer'))), deptNo varchar(10));  
Query OK, 0 rows affected (0.044 sec)
```

```
MariaDB [EmpDepartDB]> create table Department(deptNo varchar(10) primary key,  
deptName varchar(70) not null, mgrEmpNo varchar(10) not null, foreign  
key(mgrEmpNo) references Employee(empNo));  
Query OK, 0 rows affected (0.008 sec)
```

```
MariaDB [EmpDepartDB]> create table Project(projNo varchar(10) primary key,  
projName varchar(100) not null, deptNo varchar(10), foreign key(deptNo)  
references Department(deptNo));  
Query OK, 0 rows affected (0.009 sec)
```

```
MariaDB [EmpDepartDB]> create table WorksOn(empNo varchar(10), projNo  
varchar(10), dateWorked date, hoursWorked decimal(10, 2) check(hoursWorked  
between 0 and 40), primary key(empNo, projNo, dateWorked), foreign key(empNo)  
references Employee(empNo), foreign key(projNo) references Project(projNo));  
Query OK, 0 rows affected (0.009 sec)
```

##Populating tables

```
MariaDB [EmpDepartDB]> INSERT INTO Employee VALUES  
-> ('E001','John','Mensah','Accra','1990-05-10','M','Manager','D001'),  
-> ('E002','Ama','Owusu','Kumasi','1992-08-15','F','Team Leader','D001'),  
-> ('E003','Kwame','Asare','Takoradi','1995-02-20','M','Software  
Developer','D002'),  
-> ('E004','Akosua','Boateng','Cape  
Coast','1993-11-25','F','Analysts','D002'),  
-> ('E005','Yaw','Ofori','Tamale','1998-04-18','M','Software  
Developer','D003'),  
-> ('E006','Abena','Sarpong','Accra','1991-09-12','F','Manager','D003'),  
-> ('E007','Kojo','Bonsu','Tema','1994-03-30','M','Team Leader','D001'),  
-> ('E008','Efua','Nkrumah','Sunyani','1996-07-22','F','Analysts','D002'),  
-> ('E009','Kofi','Darko','Ho','1997-12-01','M','Software  
Developer','D003'),  
-> ('E010','Adjoa','Amoah','Accra','1999-06-14','F','Analysts','D001');  
Query OK, 10 rows affected (0.012 sec)  
Records: 10 Duplicates: 0 Warnings: 0
```

```

MariaDB [EmpDepartDB]> INSERT INTO Department VALUES
-> ('D001','Information Technology','E001'),
-> ('D002','Software Engineering','E002'),
-> ('D003','Research and Development','E006');
Query OK, 3 rows affected (0.005 sec)
Records: 3 Duplicates: 0 Warnings: 0

```

```

MariaDB [EmpDepartDB]> INSERT INTO Project VALUES
-> ('P001','Student Management System','D001'),
-> ('P002','Hospital Management System','D002'),
-> ('P003','E-Commerce Platform','D001'),
-> ('P004','AI Research Project','D003'),
-> ('P005','Library System','D002'),
-> ('P006','Payroll System','D001'),
-> ('P007','Mobile Banking App','D003'),
-> ('P008','Inventory System','D002'),
-> ('P009','Learning Platform','D001'),
-> ('P010','Data Analytics Tool','D003');
Query OK, 10 rows affected (0.005 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

```

MariaDB [EmpDepartDB]> INSERT INTO WorksOn VALUES
-> ('E001','P001','2026-06-01',20.00),
-> ('E002','P003','2026-06-01',15.50),
-> ('E003','P002','2026-06-02',30.00),
-> ('E004','P005','2026-06-02',25.00),
-> ('E005','P004','2026-06-03',18.00),
-> ('E006','P007','2026-06-03',22.50),
-> ('E007','P006','2026-06-04',16.00),
-> ('E008','P008','2026-06-04',19.50),
-> ('E009','P010','2026-06-05',28.00),
-> ('E010','P009','2026-06-05',14.00),
-> ('E001','P003','2026-06-06',10.00),
-> ('E003','P005','2026-06-06',12.00);
Query OK, 12 rows affected (0.005 sec)
Records: 12 Duplicates: 0 Warnings: 0

```

```

MariaDB [EmpDepartDB]> ALTER TABLE Employee
-> ADD CONSTRAINT fk_employee_department
-> FOREIGN KEY (deptNo)
-> REFERENCES Department(deptNo);
Query OK, 10 rows affected (0.017 sec)
Records: 10 Duplicates: 0 Warnings: 0

```

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```

MariaDB [EmpDepartDB]> create view FemaleManagedProjects as select p.projNo,
p.projName, p.deptNo from Project p join Department d on p.deptNo = d.deptNo
join Employee e on d.mgrEmpNo = e.empNo where e.sex = 'F' and e.emPposition =
'Manager' order by p.projNo;
Query OK, 0 rows affected (0.031 sec)

```

MariaDB [EmpDepartDB]> select * from FemaleManagedProjects;

projNo	projName	deptNo
P004	AI Research Project	D003
P007	Mobile Banking App	D003
P010	Data Analytics Tool	D003

3 rows in set (0.004 sec)

7.26

MariaDB [EmpDepartDB]> create view EmployeeProjectView as select e.empNo, e.fName, e.lName, p.projName, w.hoursWorked from Employee e join WorksOn w on e.empNo = w.empNo join Project p on p.projNo = w.projNo;
Query OK, 0 rows affected (0.030 sec)

7.27

a)MariaDB [EmpDepartDB]> select * from EmProject;

empNo	projNo	totalHours
E001	P001	20.00
E001	P003	10.00
E002	P003	15.50
E003	P002	30.00
E003	P005	12.00
E004	P005	25.00
E005	P004	18.00
E006	P007	22.50
E007	P006	16.00
E008	P008	19.50
E009	P010	28.00
E010	P009	14.00

12 rows in set (0.027 sec)

b)

Before:

MariaDB [EmpDepartDB]> select projNo from EmProject where projNo = 'SCCS';
Empty set (0.027 sec)

After:

MariaDB [EmpDepartDB]> select projNo from EmProject where projNo = 'P003';

projNo

```

| P003 |
| P003 |
+-----+
2 rows in set (0.001 sec)

```

c)

Before: Querying with

```

MariaDB [EmpDepartDB]> select count(projNo) from EmProject where empNo = 'E1';
+-----+
| count(projNo) |
+-----+
|              0 |
+-----+
1 row in set (0.026 sec)

```

After: replaced E1 with actual filled data of empNo = 'E001'

```

MariaDB [EmpDepartDB]> select count(projNo) from EmProject where
empNo = 'E001';
+-----+
| count(projNo) |
+-----+
|              2 |
+-----+
1 row in set (0.027 sec)

```

d)

```

MariaDB [EmpDepartDB]> select empNo, totalHours from EmProject group by empNo;
+-----+-----+
| empNo | totalHours |
+-----+-----+
| E001  | 20.00      |
| E002  | 15.50      |
| E003  | 30.00      |
| E004  | 25.00      |
| E005  | 18.00      |
| E006  | 22.50      |
| E007  | 16.00      |
| E008  | 19.50      |
| E009  | 28.00      |
| E010  | 14.00      |
+-----+-----+
10 rows in set (0.001 sec)

```

7.28

```

MariaDB [EmpDepartDB]> create table Part(partNo varchar(10), contract
varchar(10), partCost decimal(10,2), primary key(partNo, contract));

```

```

MariaDB [EmpDepartDB]> INSERT INTO Part VALUES
-> ('P001','C001',1200.00),
-> ('P001','C002',950.00),
-> ('P002','C001',800.00),

```

```

-> ('P002','C003',1500.00),
-> ('P003','C001',2000.00),
-> ('P003','C002',1800.00),
-> ('P004','C001',750.00),
-> ('P005','C001',1100.00),
-> ('P006','C002',500.00),
-> ('P007','C001',2500.00),
-> ('P008','C003',900.00),
-> ('P009','C002',1300.00),
-> ('P010','C001',1000.00);

```

Query OK, 13 rows affected (0.005 sec)
Records: 13 Duplicates: 0 Warnings: 0

MariaDB [EmpDepartDB]> create view ExpensiveParts(partNo) as select distinct
partNo from part where partCost>1000;
Query OK, 0 rows affected (0.031 sec)

MariaDB [EmpDepartDB]> SELECT * FROM ExpensiveParts;

```

+-----+
| partNo |
+-----+
| P001   |
| P002   |
| P003   |
| P005   |
| P007   |
| P009   |
+-----+

```

6 rows in set (0.028 sec)

Maintaining:

When the an expensive part is inserted, it is added to the view table.
In case of an update that raises the part Cost above 1000, it is added to the
view table.
deleting an expensive part must check if another contract for the same part
exists.
updating the part Cost and if it is below 1000, it must check if another
expensive contract for the same part exists.

Without table(Part):

some updates can be handled without accessing the base table.
To delete and reduce cost will generally require checking the base table
unless there is an extra summary information given.